+
INCH-POUND

MIL-C-63990C (AR)
AMENDMENT 4
15 October 1999
SUPERSEDING
AMENDMENT 3
15 SEPTEMBER 1996

MILITARY SPECIFICATION

CARTRIDGE 5.56mm, TRACER, M856

MIL-C-63990C was inactivated after 15 September 1996 For New Design.

This amendment forms a part of MIL-C-63990 (AR), dated 15 February 1994, and is approved for use by the US Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Under military specification delete the following:

"MIL-L-3150	Lubricating Oil, Preservative, Medium
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application"

* 2.1.2 Under drawings add the following:

"9349659	-	Jacket
9357840	-	Jacket, Pointed with Filler, Point"

PAGE 4

* 3.6 Delete in its entirety and substitute the following:

"3.6 Velocity. The average velocity of the cartridges, conditioned at 70 degrees plus or minus 2 degrees F, shall be 3,010 feet per second (fps) plus or minus 40 fps at 78 feet from the muzzle of the weapon. The standard deviations of the velocities shall not exceed 40 fps."

* 3.7 Delete in its entirety and substitute the following:

"3.7 Chamber pressure. The average chamber pressure of the sample cartridges, conditioned at 70 degrees plus or minus 2 degrees F, shall not exceed 58,700 psi. Neither the chamber pressure of an individual sample test cartridge nor the average chamber pressure plus three standard deviations of chamber pressure shall exceed 64,700 psi."

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* 3.8 Delete in its entirety and substitute the following:

"3.8 Port pressure. The mean port pressure minus three standard deviations shall not be less than 15,300 psi for sample cartridges conditioned at 70 degrees plus or minus 2 degrees F."

* 3.10.2 Delete in its entirety and substitute the following:

"3.10.2 Chamber pressure. The average chamber pressure shall not vary from the average chamber pressure of the sample test cartridges conditioned at 70 degrees plus or minus 2 degrees F. by more than 7,000 psi. The average chamber pressure of the sample cartridges of the same lot conditioned at 125 degrees plus or minus 2 degrees F shall not exceed 63,700 psi. Any decrease in pressure is acceptable.

* 3.10.3 Delete in its entirety and substitute the following:

"3.10.3 Port pressure. The average port pressure shall not vary by more than 1,500 psi from the average port pressure of the sample test cartridges of the same lot conditioned at 70 degrees plus or minus 2 degrees F, and shall not be less than 14,00 psi."

PAGE 8

* 4.4.1 Add the following:

"Jacket	9349659	25
Jacket, Pointed with Filler, Point	9357840	25"

* Delete 4.3.2" and substitute "4.4.2".

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* Table I Title Delete "Jacket, pointed" and substitute "Jacket"

Drawing Number Delete "9357839" and substitute "9349659".

Next Higher Assembly Delete "9342866" and substitute "9357839".

Delete "Sheet 1 of 4" and substitute "Sheet 1 of 6".

Delete "wall thickness variation (2 locations)" and substitute "wall thickness variation".

PAGE 10

* Delete "Table II" and substitute "Table I".

TABLE I Delete "Sheet 2 of 4" and substitute "Sheet 2 of 6".

Add sheet 3 of 6 included on page 3.

Add sheet 4 of 6 included on page 4.

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TABLE I Delete "Sheet 3 of 4" and substitute "Sheet 5 of 6".

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TABLE I Delete “Sheet 4 of 4” and substitute “Sheet 6 of 6”.

Under Examination or Test Column, delete “Critical” and substitute “Critical/Special”.

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4.5.6 Delete the last sentence and substitute the following:

“If a visual critical or special defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected.”

PAGE 15

Delete page 15 in its entirety and substitute new page 15 included as page 5.

TABLE I First article inspection
CLASSIFICATION OF CHARACTERISITICS

Title: Jacket, Pointed — Sheet 3 of 6 — Drawing Number 9357839

Next Higher Assembly 9342866

Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Variation in wall thickness (see note 4 on Dwg. 9357839)	25, 0-1	3.2	SMTE
Evidence of poor workmanship	25, 0-1	3.2	Visual

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TABLE I First article inspection
CLASSIFICATION OF CHARACTERISITICS

Title: Jacket, Pointed with Filler, Point

Sheet 4 of 6

Drawing Number 9357840

Next Higher Assembly 9342866

Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Length, tracer cavity (.430)	25, 0-1	3.2	SMTE
Evidence of poor workmanship	25, 0-1	3.2	Visual

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISITICS

Paragraph	Title:			
4.5.6.1	Cartridge, 5.56mm, Tracer, M856 (Inspection)	Sheet 1 of 3	Drawing Number 9342865	

Next Higher Assembly
Not Applicable

Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
CRITICAL				
1.	Perforated case (7)	100%	3.1	Gage
2.	Case split in K, L or M Location (6)	100%	3.1	Gage
3.	Weight, min. see note 1	100%	3.1	See note 2
4.	Case split in I, S or J location when it is determined that there is a potential for loss of propellant (6)	100%	3.1	Gage
SPECIAL				
a.	Primer missing (32)	100%	3.1	Visual
b.	Primer cocked (33)	100%	3.1	Visual
c.	Primer inverted (34)	100%	3.1	Visual
MAJOR				
101.	Total length	Level I	3.1	Gage
102.	Case split in I, S or J location when it is determined that there is no potential for loss of propellant (6)	100%	3.1	Gage
103.	Corrosion or stain with etching (2)	Level I	3.1	Visual
104.	Chamfer missing on head (rim) (13)	Level I	3.1	Visual
105.	Case mouth not crimped in cannelure (11)	Level I	3.1	Visual
106.	No evidence of mouth anneal (21)	Level I	3.1	Visual
107.	Draw scratch (8)	Level I	3.1	Visual
108.	Split bullet jacket (24)	Level I	3.1	Visual
109.	Loose bullet (25)	Level I	3.1	Visual
110.	Loose primer (35)	Level I	3.1	Visual

NOTES:
1. Each lightweight cartridge shall be disassembled and the propellant weighed. Any cartridge containing less than 13 grains of propellant shall be classed as a critical defect. Any cartridge containing 13 grains or more of propellant shall be classed as a major defect.

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2. One hundred percent examination for weight may be either by weighing or by measuring for propellant fill; method used must be capable of detecting a cartridge containing less than 13 grains of propellant.

PAGE 17

* 4.5.6.1 Under Minor

Characteristic 215 Delete: No waterproofing material (primer pocket joint) (37)" and substitute "Insufficient/smeared waterproofing material (primer pocket joint) (37)"

Characteristic 216 Delete "(28)" and substitute "(38)".

Characteristic 218 Delete in its entirety and substitute the following:
"Missing or improper bullet tip ID (see note 4).

Note 4 A cartridge is acceptable if it is missing up to 10% of the required bullet identification prior to linking, clipping or insertion into cartons; up to 40% following any necessary handling or orientation and after linking or clipping, or insertion into cartons".

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* 4.5.6.2 Characteristic 102 Delete acceptance criteria from "75, 7-8" and substitute "75, 6-7".

PAGE 20

* Table III Note 2 Delete first sentence in its entirety and substitute the following:

"The acceptance and retest numbers of Table IV shall be applied to the M249 results (all temperatures combined) and the M16A2 results (all temperatures combined) individually except where noted to combine them".

PAGE 22

TABLE IV Under "a, Perforation of firing pin indent in primer cup", delete the following:

"(1) M249 Machinegun 0 see 3/ 1"
and substitute:
"(1) M249 Machinegun 22 28 (see 3/) 40".

* 3c. Delete in its entirety and substitute the following:

"c. Escape of gas around primer cup

	First Sample Accept	Reject	Retest(Combined) Accept
(1) Total number of all leakers			
M16A2 Rifle	8	16	22
M249 Machinegun	17	35	67

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(2) Number of leakers that cover 50% or more of the periphery			
M16A2 Rifle	3	7	9
M249 Machinegun	3	7	9

* Add the following new defect at the end of the Table:

"7. Total defects for both M16 Rifle and M249 Machinegun combined	35	75	140"

* 4.5.9 Delete the first sentence and substitute:

"The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings specified in Table I, 4.5.6.1 and 4.5.6.2".

Note 3 Delete in its entirety and substitute the following:

"If defects in excess of the acceptance number in Table IV are noted in the first sample, a second sample consisting of double the quantity of cartridges specified under function and casualty in Table III shall be fired in both weapons. Prior to the testing of the second sample, the firing pip of the specific weapon in which the defect occurred shall be replaced with a new firing pin. If the cumulative number of defects from both the first and second sample exceeds the cumulative acceptance number in Table IV, the lot shall be rejected."

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* 6.3 Delete "Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, N.J. 07806-5000" and substitute "Commander, U.S. Army TACOM-ARDEC, ATTN: AMSTA-AR-QAC-C, Picatinny Arsenal, N.J. 07806-5000."

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* 6.6 Delete "Commander, ARDEC, ATTN: SMCAR-QAF-S, Picatinny Arsenal, N.J. 07806-5000" and substitute "Commander, U.S. Army TACOM-ARDEC, ATTN: AMSTA-AR-QAC-C, Picatinny Arsenal, N.J. 07806-5000."

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